llmXive Automated Review of Beyond the Current Observation: Evaluating Multimodal Large Language Models in Controllable Non-Markov Games

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript contains several factual inconsistencies between the narrative text and the provided data tables that undermine the accuracy of the claims.

First, in the Introduction (Section 1), the authors state: "Gemini-3.1-Pro wins all 16 head-to-head duels." However, Table 2 (tab:match_dual_image_poker_rank) presents a contradictory dataset. The table lists GPT-5.4 with 7 wins (W) and Qwen3.5-397B with 7 wins (W) in a duel setting. If Gemini won all 16 duels, the other models should have 0 wins. The text's absolute claim ("all 16") is directly refuted by the numerical data in the table it presumably describes. This suggests a confusion between a specific subset of results (perhaps text-only duels mentioned in the Appendix) and the main reported results, or a simple error in the text.

Second, the Introduction claims: "GPT-5.4 matches 69.5% of pairs on 8x10." While the number 69.5 appears in the critical elements list, Table 1 (tab:main_results) exclusively reports data for the 10x10 board size (where GPT-5.4 scores 62.3%). The 8x10 result is mentioned in the text but is absent from the main results table, leaving the claim unsupported by the visible evidence in the primary results section.

Third, the Abstract and Section 5 claim that fine-tuning Qwen3.5-9B "transfers to existing benchmarks without regression," specifically citing EMemBench and VGRPBench. While the paper mentions these benchmarks in the citations, it does not provide a table or specific numerical data showing the performance on these external benchmarks before and after fine-tuning. The claim of "no regression" is a specific negative result that requires quantitative evidence to be verified, which is currently missing from the provided text and tables.

These issues are primarily writing/accuracy errors where the text overstates or misrepresents the data presented in the tables. Correcting the text to match the tables (or vice versa) and adding the missing data points for the 8x10 board and external benchmarks is necessary to ensure claim accuracy.

Action items.

- **[writing]** Section 1 (Introduction) claims 'Gemini-3.1-Pro wins all 16 head-to-head duels,' but Table 2 (tab:match_dual_image_poker_rank) lists GPT-5.4 and Qwen3.5-397B with 7 and 7 wins respectively in a 16-game set, which is mathematically impossible if Gemini won all 16. The text contradicts the table data.
- **[writing]** Section 1 states 'GPT-5.4 matches 69.5% of pairs on 8x10,' but Table 1 (tab:main_results) only reports results for the 10x10 configuration (62.3%). The 8x10 data point is cited in the text but not present in the provided tables, making the claim unverifiable from the current evidence.
- **[writing]** Section 5 (Training) claims fine-tuning 'transfers to existing benchmarks without regression,' citing EMemBench and VGRPBench. However, the paper provides no tables or

specific numerical results for these external benchmarks to substantiate the ‘no regression’ claim, only a qualitative summary.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively illustrates the conceptual difference between Markov and Non-Markov games and outlines the test suite environments. However, the caption contains a grammatical error (‘while is non-Markov’), and the figure itself has a typo (‘Filp Back’).

Figure 2

Figure 2 effectively illustrates the two game environments (Matching Pairs and 3D Maze) with clear visual breakdowns of observations, actions, hidden states, and evaluation metrics. The layout is organized, the text is legible, and the content aligns perfectly with the provided caption.

Figure 3

The figure effectively demonstrates the performance drop as scale increases, but the x-axis labels are illegible due to overcrowding, and the right panel lacks a legend to distinguish between the Game Score and Explore % metrics.

Figure 4

The figure is a qualitative trajectory visualization that contradicts its caption, which describes a quantitative success rate plot across maze sizes. Additionally, the caption contains a broken table reference.

Figure 5

Figure 5 effectively contrasts the successful trajectory of Seed-2.0 with the failure of Kimi-K2.5, but the right panel contains confusing discrepancies between the step count title, the visualized path length, and the textual reasoning logs.

Figure 6

The figure effectively contrasts the success and failure trajectories, but the caption contains a typo regarding the maze dimensions (‘77’ instead of ‘7x7’), and the text within the small snapshot images is too small to read.

Figure 7

The figure effectively visualizes the single-player trajectory and board states, but the caption contains a typo regarding the board dimensions (‘1010’ instead of ‘10x10’) and the legend lacks a clear definition for the parenthetical score values.

Figure 8

The figure clearly visualizes the duel trajectories, but the data in the right panel (18 pairs) contradicts the caption’s claim of an 8x10 board (80 pairs), suggesting a potential error in the board size description or the plotted data.

Action items.

- [**writing**] Figure 1: The caption contains a grammatical error and missing subject in the first sentence: ‘while is non-Markov’ should read ‘while [games] are non-Markov’.
- [**writing**] Figure 1: The ‘Non-Markov’ section contains a typo ‘Filp Back’ instead of ‘Flip Back’.
- [**writing**] Figure 3: x-axis labels are illegible and appear as a dense, overlapping block of text (e.g., ‘4x4, 6x6...’) rather than distinct tick labels, making the specific board/maze sizes unreadable.
- [**science**] Figure 3: The right panel plots ‘Game Score’ and ‘Explore %’ on the same axes but lacks a legend to distinguish the two data series (solid vs. dashed lines).
- [**fatal**] Figure 4: The rendered image displays a qualitative case study of 3D Maze trajectories (Seed-2.0 vs. Kimi-K2.5) with minimaps, but the caption describes a quantitative ‘Success rate with minimap across maze sizes’ plot; the visual content does not match the caption.
- [**science**] Figure 4: The caption references ‘Tab. .’ with a missing table number, making the cross-reference unusable.
- [**writing**] Figure 5: The caption states the Kimi-K2.5 model ‘exhausts the 96-step budget,’ but the right panel’s title reads ‘Steps: 96 / 96 (failed, Eff: 0.00)’ and the trajectory visualization shows a path ending at step 73, creating ambiguity about whether the agent stopped early or the visualization is truncated.
- [**writing**] Figure 5: The text boxes for the Kimi-K2.5 model (right) contain a contradiction; Step 25 states the agent is at ‘(6,4)’ and needs to get to ‘(6,6)’, yet the trajectory map shows the agent moving away from the goal area (bottom-right) into a dead-end loop.
- [**writing**] Figure 6: The caption states the model navigates a ‘77 maze’, but the visual

grid is clearly 7x7; the caption should read '7×7' to match the image.

- [**writing**] Figure 6: The text inside the small 'Step' snapshot images (e.g., 'Step 1', 'Step 16') is illegible due to low resolution.
- [**writing**] Figure 7: The caption contains a typo '1010 noise board' which should be '10×10' to match the figure title and visual grid.
- [**writing**] Figure 7: The legend text 'GPT-5.4 (31/50)' and 'Gemini-3.1-Pro (16/50)' is ambiguous; it is unclear if these numbers represent the final score, a ratio, or a specific metric without further definition.
- [**science**] Figure 8: The caption states the board is 8x10 (80 pairs), but the 'GPT-5.4 Goes First' panel shows a cumulative total of 18 pairs, which is mathematically impossible for a single-player game on an 80-pair board (max 80) or a duel (max 40). The data appears inconsistent with the stated board size.
- [**writing**] Figure 8: The y-axis label 'Cumulative Matched Pairs' is present, but the axis ticks (0.0, 2.5, 5.0...) and the step-function nature of the data suggest discrete integer events; the decimal formatting is slightly misleading though not fatal.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific terminology that creates unnecessary barriers for non-specialist readers, particularly those in general AI or cognitive science.

In **Section 3.1 (Problem Formulation)**, the authors introduce **POMDPs** (Partially Observable Markov Decision Processes) without defining the acronym. While standard in reinforcement learning, this is a critical concept for the paper's premise and should be spelled out and briefly explained for a general audience. Similarly, the term **egocentric** (Section 3.2) is used to describe the 3D Maze view; replacing this with "first-person" or "agent-centric" would be more intuitive.

Section 5 (Training Strategy) and the **Appendix** frequently use the term **rollouts** to describe the data used for fine-tuning. This is RL jargon that obscures the meaning; "simulated game histories" or "complete game trajectories" would be more accessible. Additionally, the metric **Traj-Match** (Section 4.2, Appendix) is introduced without a clear definition of what constitutes a "match" between a model's output and the ground truth, leaving the reader to guess the calculation method.

Finally, the term **ablation** is used repeatedly in section headers (e.g., "Visual-pattern ablation," "Ask-output ablation"). While common in ML, "removal study" or "component analysis" would better convey the intent to a broader readership. The **Memory Gap** metric is well-explained, but its reliance on the undefined “oracle” concept in Equation 2 could be slightly clarified by explicitly stating that the oracle represents a model with perfect memory injection.

These changes would significantly improve the paper’s readability without sacrificing technical precision.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific terminology that creates unnecessary barriers for non-specialist readers, particularly those in general AI or cognitive science. In Section 3.1 (Problem Formulation), the authors introduce POMDPs (Partially Observable Markov Decision Processes) without defining the acronym. While standard in reinforcement learning, this is a critical concept for the paper’s premise and should be spelled out and briefly explained for a general audience. Similarly,

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a compelling benchmark for non-Markov games, but several logical inconsistencies in the presentation of metrics and results undermine the clarity of the conclusions.

First, the **Game Score (GS)** formula in Equation 1 is mathematically inconsistent with the values reported in Table 1. The formula $GS = \frac{SR + SR \times Eff + (1-SR) \times Explore}{2}$ produces a value between 0 and 1. For Gemini-3.1-Pro in the 3D Maze (13×13), the table reports $SR=50.0\%$, $Eff=62.5\%$, and $Explore=36.4\%$. Plugging these into the formula: $(0.5 + 0.5 \times 0.625 + 0.5 \times 0.364) / 2 = (0.5 + 0.3125 + 0.182) / 2 = 0.49725$, which corresponds to 49.7%. While the calculation matches the reported 49.7%, the table lists “GS%” as 49.7, implying the formula output is treated as a percentage directly. However, for GPT-5.4, $SR=20.0\%$, $Eff=75.7\%$, $Explore=32.3\%$. Calculation: $(0.2 + 0.2 \times 0.757 + 0.8 \times 0.323) / 2 = (0.2 + 0.1514 + 0.2584) / 2 = 0.3049$, or 30.5%. The values match, but the text and table headers are confusing. The primary issue is the lack of explicit normalization explanation. If the inputs are percentages (e.g., 50.0), the formula must divide by 100 or the inputs must be decimals. The table presents inputs as percentages (e.g., 50.0) but the formula implicitly treats them as decimals (0.50). This is a notational inconsistency that requires clarification to ensure the metric is reproducible and logically sound.

Second, the **Memory Gap** definition in Equation 2 and its application in Table 4 are logically flawed. The text defines $S^*(m)$ as the “score with oracle hidden state injected.” However, the data in Table 4 suggests $S^*(m)$ is actually the score achieved with the *external memory intervention* (the memory map or minimap), not the true oracle. For Qwen3.5-397B on Matching Pairs, the baseline is 38.3% and the intervention is 78.7%. The reported MemGap is 51.3%. If $S^*(m)$ were the true oracle (100%), the gap would be $1 - 0.383 = 61.7\%$. The value 51.3% is derived from $1 - (38.3/78.7)$. This means the “Memory Gap” is actually measuring the* relative improvement* provided by the external memory tool, not the gap between the model and the true oracle. This mislabeling invalidates the conclusion that "most errors are due to forgetting" (which requires comparing to the true oracle). If the gap is only 51.3% relative to the *intervention*, it implies the intervention itself is not perfect, and the remaining error could be due to planning or perception, not just memory. The authors must either redefine the metric to use the true oracle score or rephrase the conclusion to reflect that the metric measures the efficacy of the external memory tool.

Third, the **Duel Protocol** results in the Introduction and Table 2 are ambiguous. The Introduction states "Gemini-3.1-Pro wins all 16 head-to-head duels." Table 2 shows Gemini with 16 Wins, 0 Ties, 0 Losses. However, the table also lists GPT-5.4 with 7 Wins, 2 Ties, 7 Losses (50% win rate) and Qwen with 7 Wins, 1 Tie, 8 Losses (46.7% win rate). If Gemini played 16 games and won all, who did they play? If they played GPT and Qwen, the total games would be 16 (e.g., 8 vs GPT, 8 vs Qwen). But then GPT's 7 wins and Qwen's 7 wins must be against each other or against other models not listed. The text "wins all 16 matchups" is logically incomplete without specifying the opponent set. If the 16 games were against a single opponent, the other models' records are unexplained. If the 16 games were a round-robin, the total number of games would be higher. The logical structure of the tournament is unclear, making the claim "wins all" difficult to verify or interpret.

These issues are primarily notational and definitional but significantly impact the logical consistency of the paper's claims. The metrics are not defined in a way that matches their calculation, and the experimental setup is described ambiguously.

Action items.

- [**writing**] The paper presents a compelling benchmark for non-Markov games, but several logical inconsistencies in the presentation of metrics and results undermine the clarity of the conclusions. First, the Game Score (GS) formula in Equation 1 is mathematically inconsistent with the values reported in Table 1. The formula $GS = \frac{SR + SR \times Eff + (1-SR) \times Explore}{2}$ produces a value between 0 and 1. For Gemini-3.1-Pro in the 3D Maze (13×13), the table reports $SR=50.0\%$, $Eff=62.5

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend slightly beyond the strict evidence provided in the text and tables, primarily regarding the scope of transfer learning results and the causal attribution of failure modes.

First, the Abstract and Introduction assert that fine-tuning on the proposed benchmark "transfers to existing benchmarks without regression." While Section 5 and Table 3 demonstrate positive gains on EMemBench and VGRPBench, the manuscript does not explicitly list or report results on the "general multimodal tasks" used to verify the "no regression" claim. Without a table or explicit statement confirming that performance on standard benchmarks (e.g., MMBench, MMMU) remained stable or improved, this is an over-claim of the transferability study's scope. The authors should either provide the specific regression data or soften the claim to "transfers to specific episodic memory benchmarks with gains."

Second, the claim that "Memory Gap analysis attributes most errors to forgetting" (Abstract) is a strong causal statement. The Memory Gap metric (Eq. 2) quantifies the performance difference between the model and an oracle with perfect hidden state. While a large gap implies the model is not utilizing past information, it conflates "forgetting" (failure to retain) with "reasoning failure" (failure to use retained information to plan). The paper does not provide a separate ablation that isolates retention from reasoning (e.g., by providing the history but masking the reasoning steps). Therefore, attributing the errors *specifically* to forgetting is an over-interpretation of the metric; the data supports that the models fail to reconstruct the belief state, but the root cause (memory vs. reasoning) is not fully disentangled.

Finally, the statement "Gemini-3.1-Pro wins all 16 head-to-head duels" in the Introduction is slightly ambiguous given Table 2. The table shows a 16-0-0 record, but the text in Section 5.1 mentions GPT-5.4 and Qwen3.5-397B tying at ~47-50% win rate. If the 16 games were a mix of matchups, the phrasing "wins all 16" could be misread as a sweep against every opponent individually in a round-robin, whereas the table aggregates the results. Precision in describing the duel protocol's outcome is required to avoid over-simplifying the competitive landscape.

Action items.

- [**writing**] The abstract claims fine-tuning transfers to benchmarks 'without regression,' but Section 5 only reports gains on EMemBench/VGRPBench. No data is shown for general multimodal tasks to verify the 'no regression' claim, making this an over-claim of scope.
- [**science**] The abstract states errors are 'mostly forgetting' based on Memory Gap. However, the metric only measures the gap to an oracle, conflating forgetting with reasoning failure. The paper lacks ablation to isolate retention from planning, so this causal attribution is unsupported.
- [**writing**] The introduction claims Gemini-3.1-Pro 'wins all 16 head-to-head duels,' but Table 2 aggregates results against multiple opponents. The phrasing implies a specific sweep that may mislead readers about the exact matchup distribution; precise qualification is needed.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a benchmark for evaluating Multimodal Large Language Models (MLLMs) in controllable non-Markov games, specifically focusing on memory and belief-state reconstruction. From a safety and ethics perspective, the work is sound and poses no immediate risks.

The authors explicitly address ethical considerations in the “Ethics Statement” (Section 7) and “Potential Risks” (Section 8). They correctly identify that the visual observations are synthetically generated, eliminating concerns regarding human-subject data, privacy violations, or the need for IRB/IACUC approval. The environments (Matching Pairs and 3D Maze) are abstract and do not simulate real-world harmful scenarios, reducing the risk of dual-use for malicious training or social engineering.

The “Potential Risks” section appropriately frames the benchmark as a diagnostic tool for research evaluation rather than a metric for deployment in safety-critical applications, mitigating the risk of over-interpretation. The fine-tuning experiments utilize optimal rollouts from a simulator, which are synthetic and do not introduce biased or harmful human data.

No conflicts of interest are apparent in the provided text. The data and code availability are handled via external links, which is standard practice and does not raise privacy concerns. The paper does not contain any content that could be interpreted as promoting harm, hate speech, or dangerous activities. The scope of the research is strictly limited to evaluating model capabilities in a controlled, synthetic environment.

Therefore, the paper meets the safety and ethics criteria for acceptance.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel benchmark for non-Markov games, but the scientific evidence supporting the central claims requires clarification on statistical rigor and data consistency.

First, there is a critical contradiction in the **Duel Protocol** results. The narrative in Section 4.1 states, “Gemini-3.1-Pro wins all 16 matchups.” However, Table 2 explicitly lists GPT-5.4 with 7 wins and 7 losses (50% win rate) and Qwen3.5-397B with 7 wins and 8 losses. If Gemini won *all* 16, the opponents should have 0 wins. This discrepancy undermines the claim of total dominance and suggests a potential error in either the data aggregation or the textual summary.

Second, the **Memory Gap** metric (Eq. 2) is central to the claim that “most errors are due to forgetting.” The metric is defined as $1 - S(m)/S^*(m)$. The paper describes $S^*(m)$ as the score with “oracle hidden state injected.” It is unclear if this oracle is a perfect solver (achieving 100% score) or a model with perfect memory but the same reasoning limitations as the baseline. If the oracle is not a perfect solver, $S^*(m) < 100\%$, and the “gap” conflates memory failure with reasoning failure. Without defining the oracle’s upper bound, the attribution of errors solely to “forgetting” is not fully supported.

Third, the **fine-tuning results** (Section 5, Table 4) lack statistical robustness. The improvement from 0.0% to 29.5% on Matching Pairs and 0.0% to 10.0% on 3D Maze is reported for a single training run. The paper mentions evaluation aggregates over 5 environment seeds, but it does not report the variance or standard deviation of the fine-tuned model's performance across these seeds. Given the high variance often seen in LLM evaluation, a single point estimate is insufficient to claim a robust transfer of capability.

Finally, the **sample sizes** for the 3D Maze success rates are extremely low (e.g., "1/5", "2/5" in Appendix tables). While the paper acknowledges this in the "Limitations" section, the main text presents these low-probability events as definitive performance metrics (e.g., "0.0% SR"). The evidence for the "bottleneck" being visual recognition (Section 4.2) is strong, but the evidence for the specific efficacy of the fine-tuning strategy is weak due to the lack of error bars and the small number of successful episodes.

Action items.

- [**science**] The 'Duel Protocol' results (Tab. 2) claim Gemini-3.1-Pro won all 16 matchups against GPT-5.4 and Qwen3.5-397B. However, the table lists GPT-5.4 with 7 wins and 7 losses (50% win rate) and Qwen3.5-397B with 7 wins and 8 losses. This is a direct contradiction between the narrative text and the data table that must be resolved to validate the claim of total dominance.
- [**science**] The Memory Gap metric (Eq. 2) relies on an oracle score $S^*(m)$. The paper states the oracle injects 'true hidden state,' but does not specify if the oracle is a perfect solver (100% success) or a model with perfect memory but limited reasoning. If the oracle is not perfect, the denominator $S^*(m)$ is ambiguous, making the 'gap' a measure of both memory and reasoning deficits rather than pure forgetting.
- [**science**] The fine-tuning study (Sec. 5, Tab. 4) reports a single run for Qwen3.5-9B with 'rmix32k' data. No standard deviation or confidence intervals are provided for the 29.5% score or the 10.0% success rate. Given the stochastic nature of RLHF/SFT and the small sample size of the evaluation (implied by the integer success counts in other tables), the statistical significance of the transfer claim is unverified.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the manuscript requires clarification regarding sample sizes, variance reporting, and the consistency of reported metrics.

First, there is a significant inconsistency in the **Duel Protocol** results presented in Section 5.1 and Table 2. The table lists 16 total matchups (W+T+L) with Gemini-3.1-Pro winning 16 (100% win rate). However, the accompanying text in Section 5.1 states: "GPT-5.4 and Qwen3.5-397B tie at ~47-50% win rate." This is mathematically impossible if the total sample is only 16 games where one model won every single game. It is unclear if the "16" refers to a subset of games, if the win rates are aggregated across different board sizes/seeds not fully detailed in the table, or if there is a reporting error. The sample size (N) and the exact aggregation logic for the win rates must be explicitly defined to ensure reproducibility.

Second, the **Memory Gap** metric (Eq. 2) is a critical diagnostic tool in this paper, yet the statistical uncertainty surrounding these values is not addressed. The authors report specific MemGap values (e.g., 51.3% for Qwen on Matching Pairs, 40.8% for 3D Maze) derived from baseline and oracle scores. However, the raw success rates in the ablation studies (e.g., Table 5, "Maze-size sweep") show high variance (e.g., "1/5" or "2/5" successes). Without reporting standard errors, confidence intervals, or the results of significance tests (e.g., t-tests or bootstrap resampling) for the Memory Gap, it is difficult to determine if the observed differences between models or conditions are statistically significant or due to random variance in the small sample sizes.

Third, the **fine-tuning results** in Table 4 lack measures of dispersion. The paper claims a jump from 0.0% to 29.5% for Qwen3.5-9B on Matching Pairs. While the appendix mentions evaluation over "5 environment seeds," the main results table presents only point estimates. Given the stochastic nature of the game environment and the small number of seeds (N=5), the standard deviation of these scores is likely non-negligible. Reporting the mean $\pm$ standard deviation (or standard error) is essential to validate the claim that the improvement is robust and not an artifact of a specific seed configuration.

Finally, the **scale-sweep analysis** (Section 5.2) notes a sharp drop in performance as grid size increases. While the trend is clear, the paper does not quantify the rate of decay or test for a significant interaction between model size and grid complexity. A regression analysis or a formal test of the slope of the performance curve would strengthen the claim that "performance collapses as latent state grows."

In summary, while the experimental design is sound, the statistical reporting needs to be tightened to include variance estimates, clarify sample sizes for aggregated metrics, and ensure internal consistency between tables and text.

Action items.

- [**science**] The 'Duel Protocol' results (Table 2) report 16 total matchups with a 100% win rate for Gemini-3.1-Pro. However, the text states 'GPT-5.4 and Qwen3.5-397B tie at ~47-50% win rate' in the same section, which is mathematically inconsistent with a 16-game sample where one model won all games. Clarify the sample size and aggregation method for the duel statistics.
- [**science**] The 'Memory Gap' metric (Eq. 2) is defined as a percentage difference from an oracle score. The paper reports specific MemGap values (e.g., 51.3%, 40.8%) but does not provide confidence intervals or standard errors for these derived metrics, despite the high variance observed in the raw success rates (e.g., 1/5 vs 4/5 in Table 5). Statistical significance of these gaps is not established.
- [**science**] In the fine-tuning section (Table 4), the improvement of Qwen3.5-9B from 0.0% to 29.5% on Matching Pairs is reported. The text mentions evaluation over '5 environment seeds' in the appendix, but the main table presents single-point estimates without error bars

or standard deviations. Given the stochastic nature of the game and the small sample size implied by the 1/5 success rates elsewhere, the statistical robustness of the 29.5% figure is unclear.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript is generally well-written, with a clear structure and logical flow that effectively communicates the benchmark design and experimental results. The abstract and introduction successfully set the stage for the problem of non-Markov evaluation in MLLMs. However, there are specific instances where precision in language and data reporting could be improved to prevent minor ambiguities for the reader.

In Section 5.1, the summary of the Duel results states that GPT-5.4 and Qwen3.5-397B “tie at ~47-50% win rate.” While the range is technically correct, the phrasing “tie” suggests statistical equivalence, whereas Table 2 explicitly lists 50.0% for GPT-5.4 and 46.7% for Qwen3.5-397B. A more precise phrasing, such as “perform similarly with win rates of 50.0% and 46.7% respectively,” would better reflect the data without implying a formal tie.

Additionally, in Section 5.2, the claim that “Text-only solves Matching Pairs (100%) and 3D Maze (GS ~70%)” conflates two different performance metrics under the verb “solves.” While 100% implies a complete solution, a Game Score of ~70% does not necessarily constitute a “solution” in the same absolute sense. Clarifying this distinction (e.g., “achieves perfect scores on Matching Pairs and ~70% GS on 3D Maze”) would enhance the accuracy of the claim.

Finally, the Appendix uses the abbreviation “Tabs.” when referring to multiple tables. For consistency with the main body of the paper, which spells out “Table” in text and captions, the authors should consider using the full word “Tables” to maintain a uniform academic tone throughout the document. These are minor stylistic and precision issues that do not detract significantly from the overall readability but are worth addressing for a polished final version.

Action items.

- **[writing]** In Section 5 (Experiments), the text states ‘GPT-5.4 and Qwen3.5-397B tie at ~47-50% win rate’ in the Duel setting. However, Table 2 shows GPT-5.4 at 50.0% and Qwen3.5-397B at 46.7%. The text should clarify that these are distinct values or adjust the phrasing to avoid implying a statistical tie where the data shows a difference.
- **[writing]** In Section 5.2 (Diagnostic Analysis), the sentence ‘Text-only solves Matching Pairs (100%) and 3D Maze (GS ~70%)’ is ambiguous. It is unclear if ‘solves’ implies perfect performance or merely high performance. Given the context of ‘100%’, it likely means perfect, but ‘GS ~70%’ is not a ‘solution’ in the same sense. Rephrase for precision, e.g., ‘Text-only inputs achieve 100% on Matching Pairs and ~70% GS on 3D Maze.’

- **[writing]** In the Appendix (More Analysis), the phrase 'Tabs.~\ref{tab:rq1_board_size_match} and~\ref{tab:rq1_board_size_maze} report the per-size numbers' uses 'Tabs.' as an abbreviation for 'Tables'. While common in some contexts, standard academic prose usually spells out 'Tables' or uses 'Table' if referring to a single one. Ensure consistency with the main text style, which spells out 'Table' in captions and text.